

10 - Step Design of Post-Tensioned Floors



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301 Mission Street
San Francisco, California
High Seismic Force Region



Four Seasons Hotel; Florida
High Wind Force Region



Residential/Office Post-
Tensioned Building in Dubai



Column supported multistory building

Two-way flat slab construction



Multi-level parking structures

One-way beam and slab design

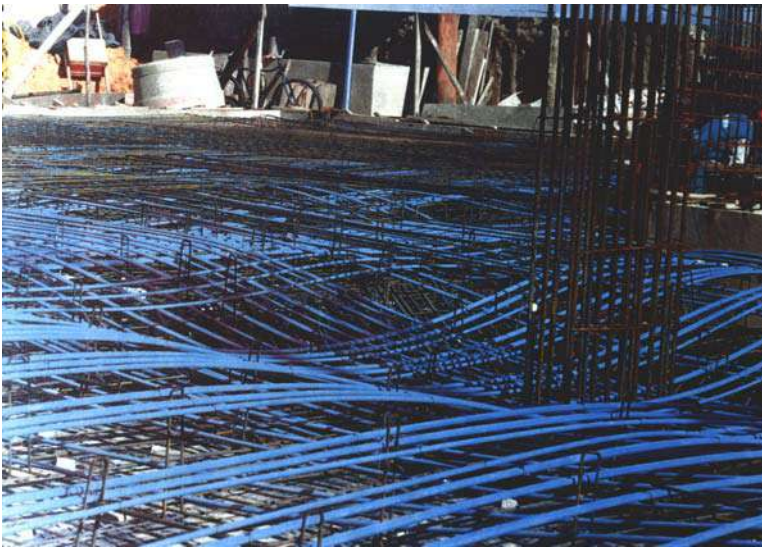


Hybrid Construction
Post-tensioned podium slab supporting light framed structure above

Santana Row; San Jose, California



Post-Tensioned ground supported slab (SOG) is the largest application of post-tensioning in USA



Fortaleza, Brazil

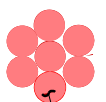
POST-TENSIONED MAT FOUNDATION USING UNBONDED TENDONS



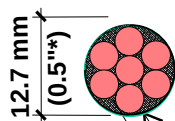
KSA

POST-TENSIONED MAT FOUNDATION USING GROUTED TENDONS

Post-Tensioning Systems Unbonded System



WIRE



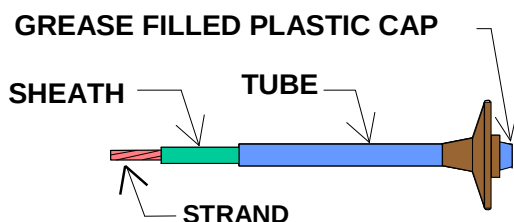
CORROSION
INHIBITING
COATING

PLASTIC
SHEATHING

NOTE: * NOMINAL DIAMETER

(a) STRAND

(b) TENDON

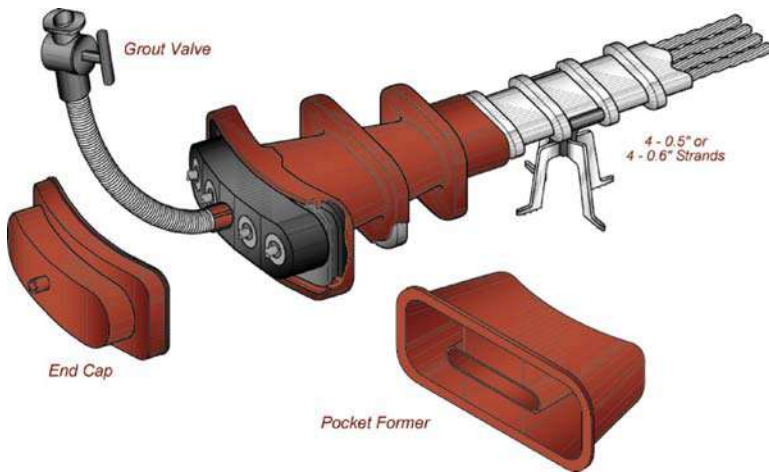


(c) ANCHORAGE ASSEMBLY

Example of a Floor System using the Unbonded Post-tensioning System

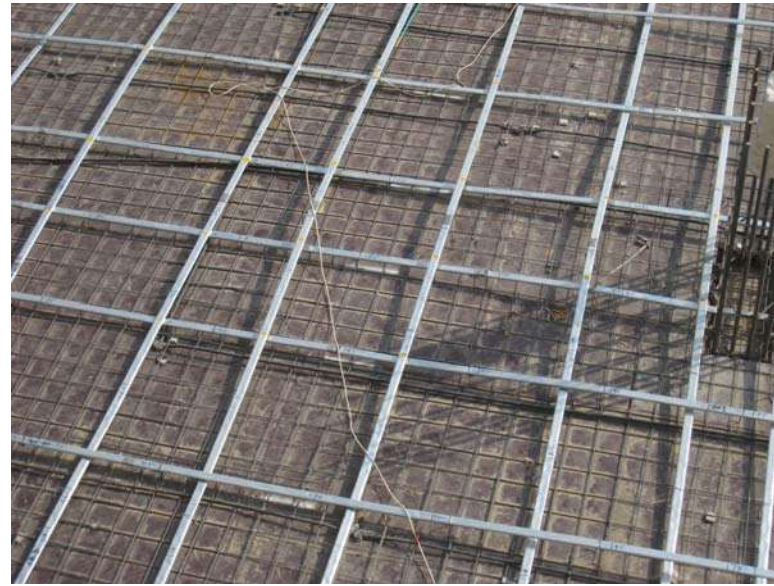


Post-Tensioning Systems Grouted System



An example of a grouted system hardware with flat duct

Example of a Floor System Reinforced with Grouted Post-Tensioning System



Preliminary Considerations Design of Post-Tensioned Floors

- ❖ **Dimensions** (sizing)
 - Optimum spans; optimum thickness
- ❖ **Structural system**
 - One-way/**two-way**; slab band/beam
- ❖ **Boundary conditions; connections**
 - Service performance; strength condition

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Preliminary Considerations Design of Post-Tensioned Floors

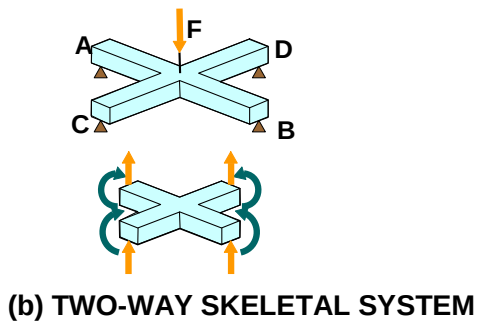
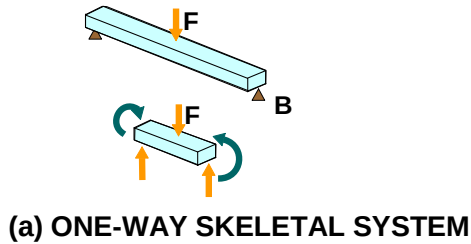
- ❑ **Dimensions** (sizing)
 - Optimum spans; optimum thickness
- ❖ An optimum design from standpoint of **material usage** is one in which the **reinforcement** determined for “service condition” is **used in its entirety** for “strength condition.”
- ❖ PT **amount** in service condition is governed mostly by:
 - Hypothetical **tensile stresses**
 - Tendon **spacing**
- ❖ Optimum spans: between 25 ft – 30 ft
- ❖ Span/thickness ratios
 - 40 for interior
 - 35 for exterior with no overhang

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Preliminary Considerations Design of Post-Tensioned Floors

❖ One-way and two-way systems

- Skeletal Systems **share** load
- Slab systems **may not share** load, depending on **reinforcement layout**

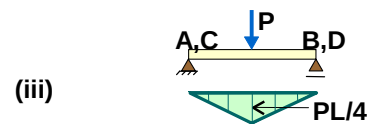
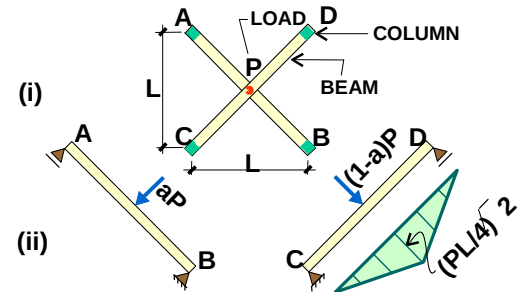


Skeletal System

Preliminary Considerations Design of Post-Tensioned Floors

❖ One-way and two-way systems

- Skeletal Systems **share** load
- Slab systems **may not share** load depending on reinforcement layout



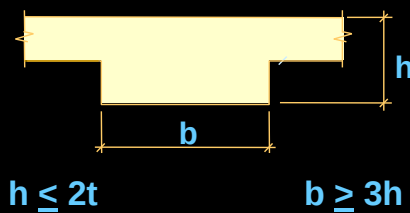
(b) BEAMS ON FOUR COLUMNS

Capacity required is for **100% of load** in each direction, if reinforcement is placed along AD, and AC

Preliminary Considerations Design of Post-Tensioned Floors

❖ Structural system

- One-way/two-way; **slab band**



$$h \leq 2t$$

$$b \geq 3h$$



SLAB BAND

Slab band is treated as part of a two-way system. One-way shear design provisions meant for beams do not apply to slab bands

Preliminary Considerations Design of Post-Tensioned Floors

❖ Selection of boundary conditions; connections

- Service performance
- Strength performance

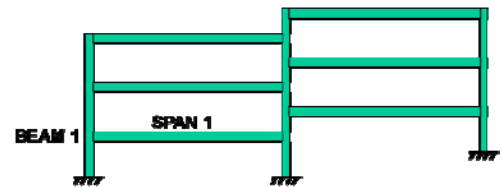
Detailing for service performance, such as the one shown below is to mitigate cracking from shortening of slab



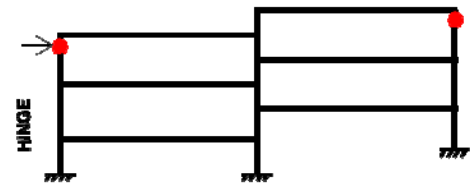
Preliminary Considerations Design of Post-Tensioned Floors

Assumption of releases at connections, or reduced stiffness for selected members is made **prior to analysis** to achieve a more economical design.

In the following, the assignment of **reduced stiffness** for the uppermost columns, or **hinge assumption** at connection is not uncommon



(a) FRAME ELEVATION

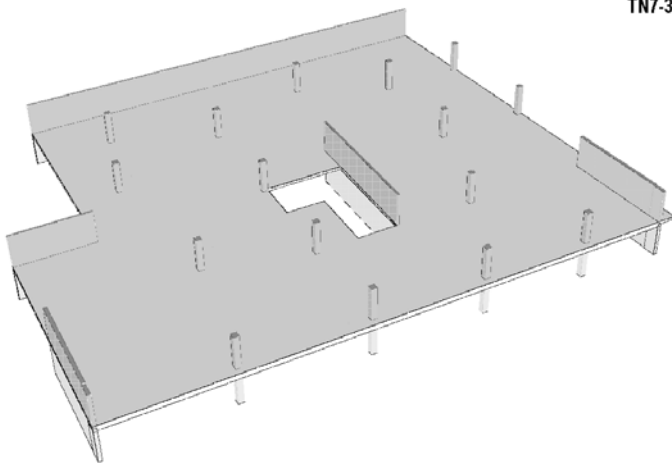


(b) STRUCTURAL MODEL

Preliminary Considerations Design of Post-Tensioned Floors

❖ Selection of load path for two-way systems – **Design Strips**

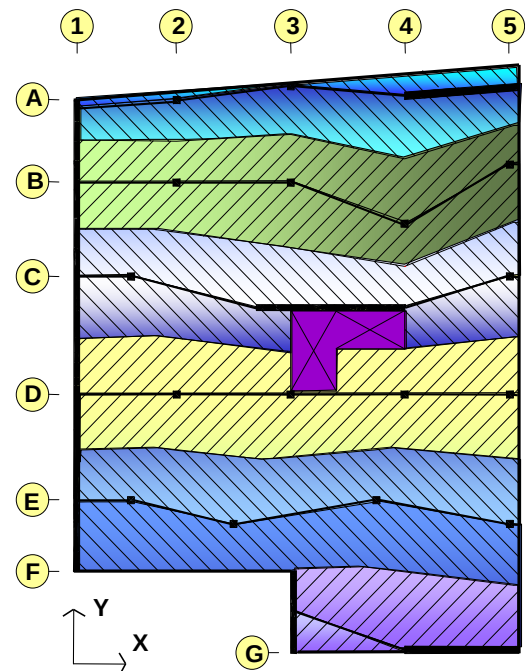
TN7-369



Subdivide the structure into design strips in two orthogonal directions
(Nahid slab)

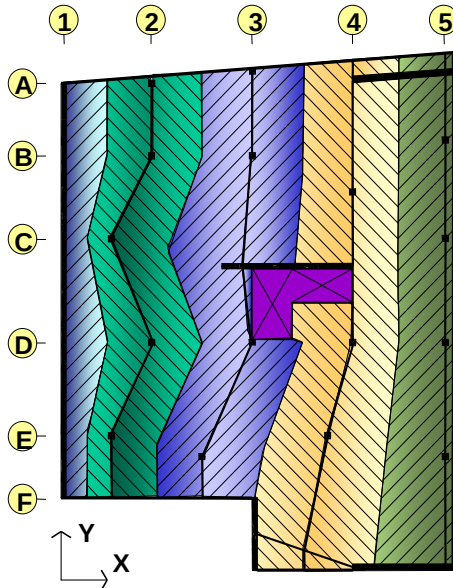
Preliminary Considerations Design of Post-Tensioned Floors

Subdivide the floor along line of columns into **design strips**



Preliminary Considerations Design of Post-Tensioned Floors

Subdivide floor along support lines in design strips in the **orthogonal direction**

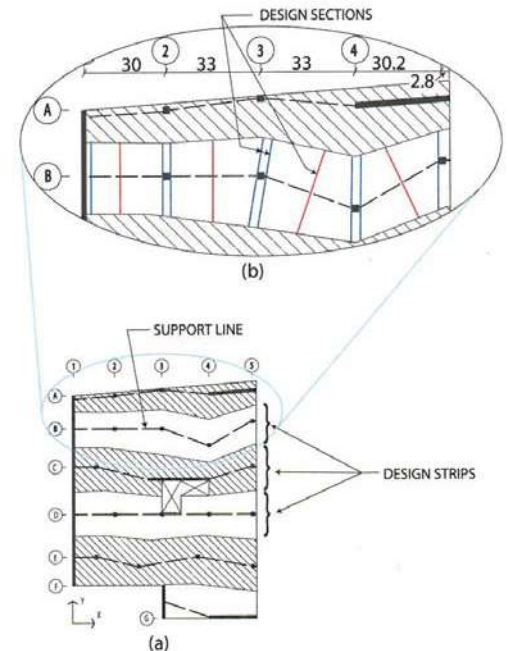


An important aspect subdivision of slab into design strips is that **every point** of the slab should be **covered by a design strip**. No portion of the slab should be left unassigned.

Preliminary Considerations Design of Post-Tensioned Floors

Design sections

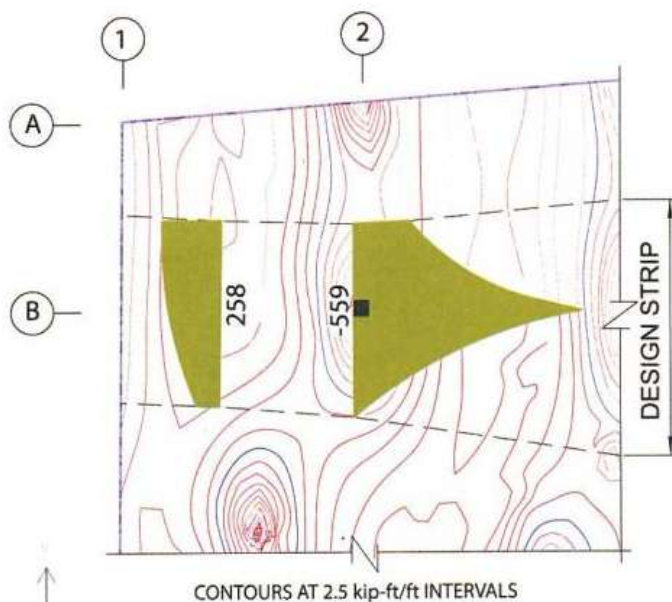
- Design sections extend over the **entire design strip** and are considered at critical locations, such as face of support and mid-span



Preliminary Considerations Design of Post-Tensioned Floors

❖ Design values

- Actions, such as moments at each design section are reduced to a **“single” representative value** to be used for design



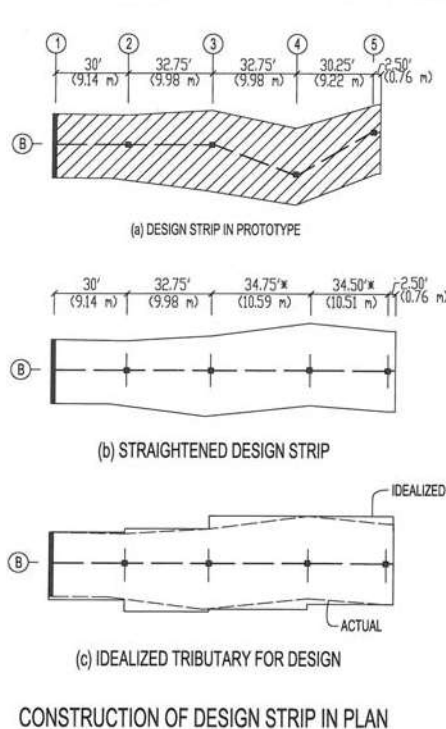
559 k-ft is the area (total) of bending moment at face of support

10- Steps Design of Post Tensioned Floors

1. **Geometry** and Structural System
2. **Material** Properties
3. **Loads**
4. Design **Parameters**
5. Actions due to Dead and Live Loads
6. Post-Tensioning
7. Code Check for **Serviceability**
8. Code Check for **Strength**
9. Check for **Transfer of Prestressing**
10. **Detailing**

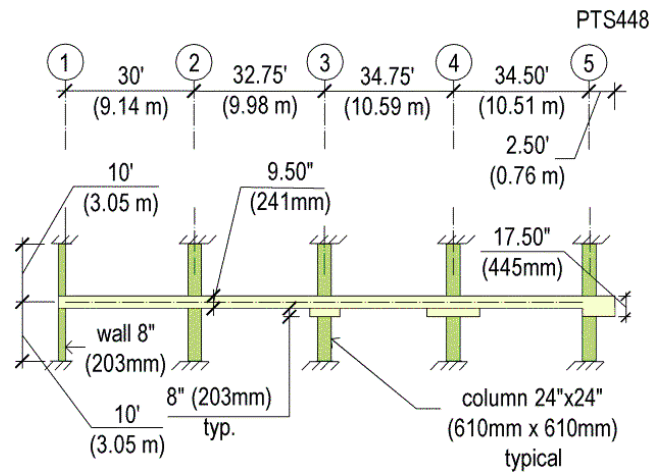
Step 1 Geometry and Structural System

- ❖ Select design strip and **Idealize**
 - **Extract**; **straighten** the support line; **square** the boundary



Step 1 Geometry and Structural System

- Select design strip and **Idealize**
 - Model the slab frame with a row of supports above and below. This represents an upper level of multi-story concrete frame.
 - Assume **rotational fixity** at the far ends;
 - Assume **roller** support at the far ends



Step 2 Material Properties

Concrete

- Weight 150 pcf
- 28 day cylinder 4000 – 6000 psi
- Elastic modulus 3,605 – 5,700 ksi
- Long-term deflection factor 2

❖ Non-Prestressed reinforcement

- f_y 60 ksi
- Elastic modulus 29,000 ksi

❖ Prestressing

- Strand diameter 0.5 in
- Strand area 0.153 in²
- Ultimate strength 270 ksi
- Effective stress 175 ksi
- Elastic modulus 28,000 ksi

Step 3 Dead and Live Loads

❖ Selfweight

- Based on member volume

❖ Superimposed dead load

- Min (partitions) 20 psf

❖ Live load

- Residential 40 psf
- Office 50 psf
- Shopping mall 75 psf
- Parking structure 40 psf

❖ Lateral loads

- Wind
- Earthquake

Step 4 Design Parameters

❖ Applicable code

- ACI 318-14
- IBC 2015
- Local codes, such as California Building Code (CBC 2011)

☐ Cover for protection against corrosion

❖ Cover to rebar

- Not exposed to weather 0.75 in
- Exposed to weather 2.00 in

❖ Cover to tendon

- Not exposed to weather 0.75 in.
- Exposed to weather 1 in.

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Step 4 Design Parameters

☐ Cover for fire resistivity

❖ Identify “restrained” and “unrestrained panels.”

Restrained or Unrestrained	Aggregate Type	Cover Thickness, in. For Fire Endurance of				
		1 hr	1.5 hr	2 hr	3 hr	4 hr
Unrestrained	Carbonate	-	-	1.50	2.00	-
	Siliceous	-	-	1.50	2.00	-
	Lightweight	-	-	1.50	2.00	-
Restrained	Carbonate	-	-	0.75	1.00	1.25
	Siliceous	-	-	0.75	1.00	1.25
	Lightweight	-	-	0.75	1.00	1.25

❖ For 2-hour fire resistivity

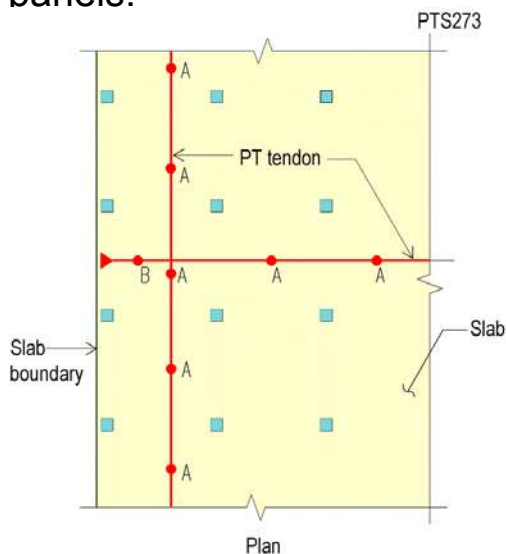
- Restrained 0.75 in.
- Unrestrained 1.50 in

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Step 4 Design Parameters

☐ Cover for fire resistivity

❖ Identify “restrained” and “unrestrained panels.”



A = bottom cover for restrained condition
B = bottom cover for unrestrained condition

Bottom Cover to Tendon for Fire Resistivity

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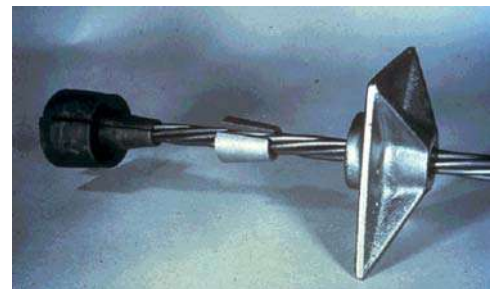
Step 4 Design Parameters

❖ Select post-tensioning system

- For **corrosive** environment use “encapsulated system.”



- For **non-corrosive** environment, regular hardware may be used



Step 4 Design Parameters

□ Allowable stresses for two-way systems

❖ Service condition

➤ Total (frequent) load case

- Tension ----- $6\sqrt{f'_c}$
- Compression ----- $0.6f'_c$

➤ Sustained (quasi permanent) load case

- Tension ----- $6\sqrt{f'_c}$
- Compression ----- $0.45f'_c$

❖ Initial condition

- Tension ----- $0.6\sqrt{f'_c}$
- Compression ----- $0.45f'_c$

Step 4 Design Parameters

□ Allowable deflections (ACI 318)

❖ For **visual impact** use total deflection

- **Span/240**
- Use camber, if necessary

❖ **Total** deflection subsequent to installation of members that are likely to **be damaged**

- **Span/360**

❖ **Immediate** deflection due to **live load**

- **Span/480**

❖ **Long-term** deflection magnifier 2. This brings the total long-term deflection to 3,

Step 5 Actions due to Dead and Live Loads

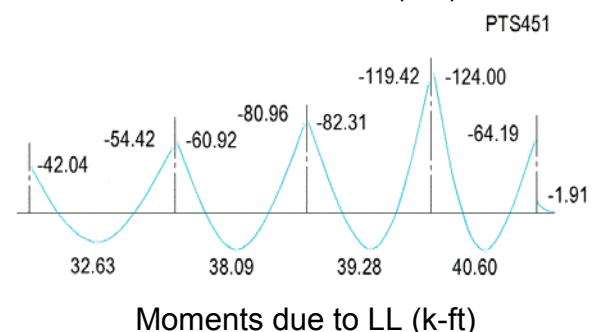
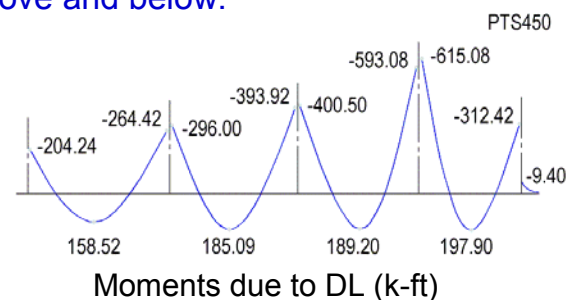
□ Analyze the design strip as a single level frame structure with one row of supports above and below, using

- ❖ In-house **simple frame** program (Simple Frame Method; SFM); or
- ❖ in-house **Equivalent Frame** Program (EFM);
- ❖ Specialty **commercial software**

- ❖ All the three options yield **safe designs**. But, each will give a different amount of reinforcement.
- ❖ The **EFM** is suggested by **ACI-318**. To some extent, it accounts for biaxial action of the prototype structure.
- ❖ 3D **FEM** software can improve **optimization**.

Step 5 Actions due to Dead and Live Loads

□ Analyze the design strip as a single level frame structure with one row of supports above and below.



Step 6 Post-Tensioning

Selection of **design parameters**

- ❖ Selection of **PT force and profile**
- ❖ **Effective force** vs tendon selection option
 - Force selection option
- ❖ Calculation of **balanced loads**; adjustments for **percentage** of load **balanced**
- ❖ **Calculation of actions** due to balanced loads

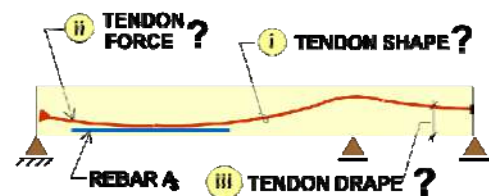
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Step 6 Post-Tensioning

- ❖ Selection of **PT force and profile**
 - **Two entry value** selections must be made to initiate the computations. Select **precompression** and **% of DL** to balance



(a) NONPRESTRESSED MEMBER



(b) POST-TENSIONED MEMBER

ADAPT

Step 6 Post-Tensioning

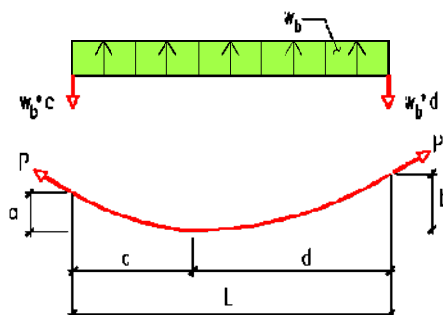
Selection of **design parameters**

- Select average precompression **150 psi**
- Target to balance **60% of DL**

- ❖ Selection of **PT force and profile**
 - Assume **simple parabola** mapped within the bounds of top and bottom covers

Force diagram of simple parabola

PTS07 07731



$$c/L = [\sqrt{a/b} / (1 + \sqrt{a/b})]$$

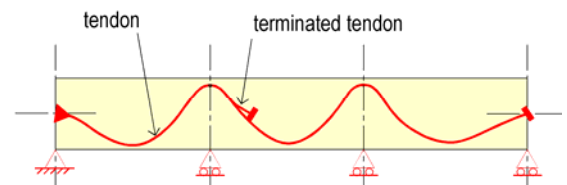
$$w_b = 2P * c/c^2$$

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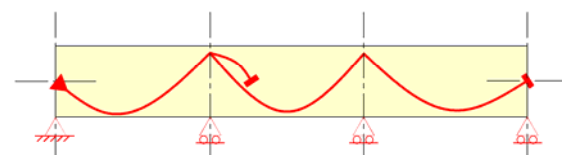
Step 6 Post-Tensioning

- Assume **simple parabola** for **hand calculation**

PTS488



(a) Tendon profiles in prototype



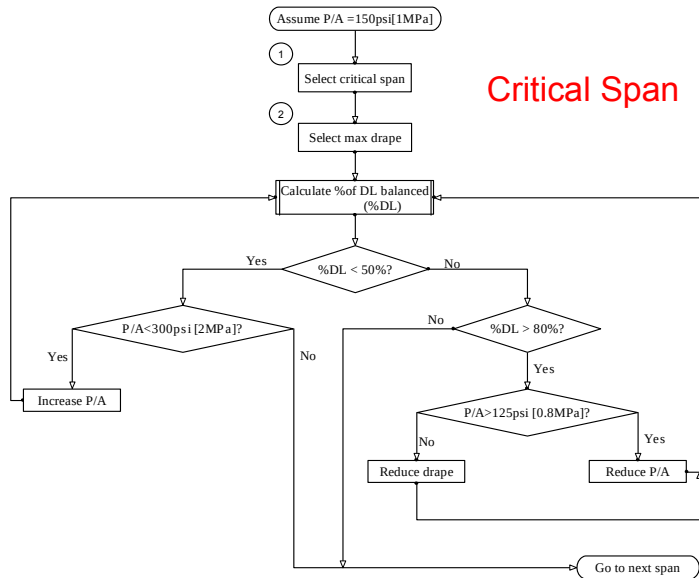
(b) Tendon profiles for hand calculation

Tendon Profiles in Prototype and Hand Calculation

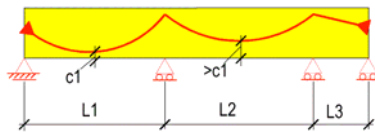
STEP 6 Post-Tensioning

- ❖ **Flow chart** for Calculation of balanced loads and adjustment for % of DL balanced

F121_ACI_PT_2-way_082012



Critical Span



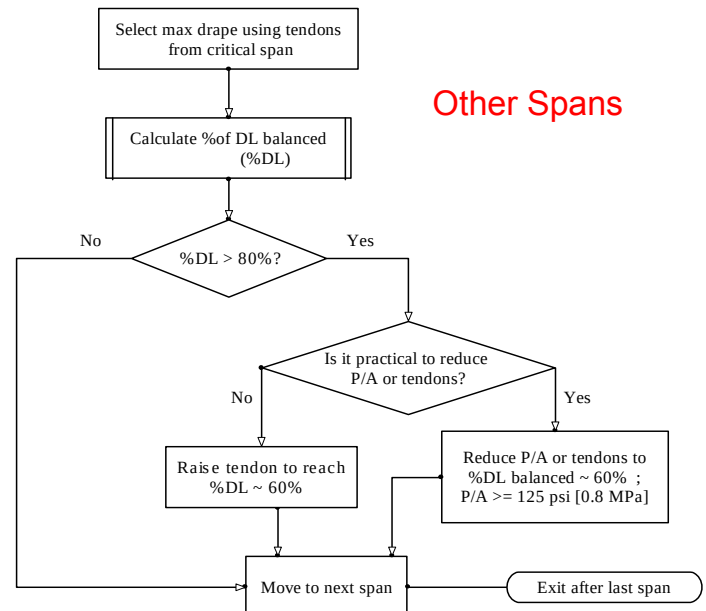
PTS526

Element with widely different spans

STEP 6 Post-Tensioning

- ❖ **Flow chart** for calculation of balanced loads and adjustment for % of DL balanced

F121_ACI_2-way_PT_force_082012



Other Spans

STEP 6 Post-Tensioning

- ❖ **Effective force** per strand
 - **Unbonded:** $0.153 \times 175 = 26.77$ k/tendon
 - Bonded:** $0.153 \times 160 = 24.48$ k/tendon

- ❖ Estimate initial force for each span using the assumed **150 psi** precompression

Span 1

$$\text{Force} = (26.77 \times 12 \times 9.5) \times 150 / 1000 = 448.88 \text{ k}$$

$$\text{No of tendons} = 448.88 / 26.77 = 16.77$$

assume 17 strands

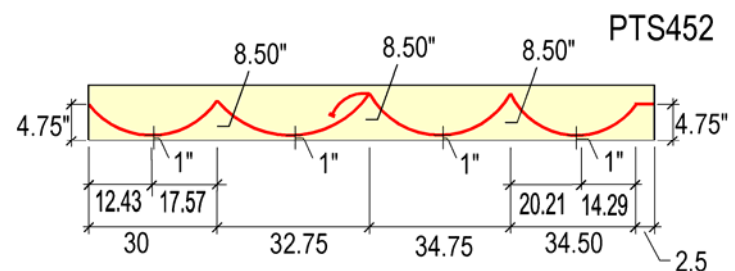
Similarly the required strands are:

	required	assumed
Span 1	17	20
Span 2	20	20
Span 3	23	23
Span 4	22	23

	required	assumed
Span 1	17	20
Span 2	20	20
Span 3	23	23
Span 4	22	23

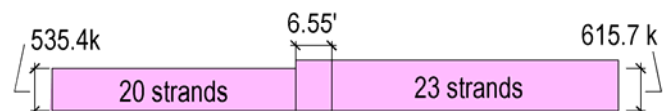
STEP 6 Post-Tensioning

- ❖ **Tendon selection**



PTS452

(a) Assumed tendon profile (ft UNO)

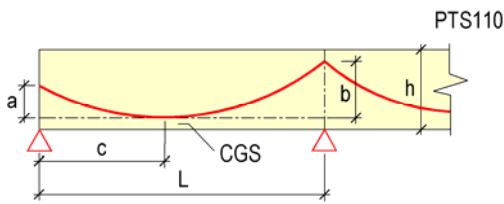


(b) Force diagram

Post Tensioning Profile and Force

STEP 6 Post-Tensioning

- ❖ Calculation of **balanced loads**
- ❖ Span 1



$$\begin{aligned}
 a &= 3.73 \text{ in} \\
 b &= 7.5 \text{ in} \\
 L &= 30.00 \text{ ft} \\
 c &= \{[3.75/7.5]^{0.5}/[1 + (3.75/7.5)^{0.5}]\} * 30 = \\
 &= 12.43 \text{ ft} \\
 P &= 26.77 \text{ k/strand} \\
 w_b &= 2 P * a / c^2 = (2 * 26.77 * 3.75 / 12) / 12.43^2 \\
 &= 0.108 \text{ klf per tendon} \\
 \text{Total uplift for 20 strands} &= 20 * 0.108 = 2.16 \text{ klf} \\
 \% \text{ uplift} &= (2.16 / 3.826) * 100 = 56\% < 60\% \\
 &\text{but considered acceptable}
 \end{aligned}$$

STEP 6 Post-Tensioning

- ❖ Calculation of **balanced loads**
 - Span 2
 - Span 2 has 20 continuous tendons and 3 tendons from span 3 extending to span 2

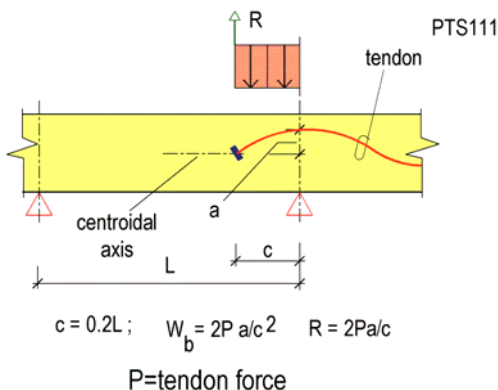
- ❖ **Balanced loads** of span 2 consist of:
 - Lateral forced from **continuous tendons**
 - Lateral force from **terminated tendons**
 - **Moments from change in centroid** of member

Continuous strands:

$$\begin{aligned}
 w_b &= 8 P * a / L^2 \\
 &= (8 * 26.77 * 7.5 / 12) / 32.75^2 = 0.125 \text{ klf} \\
 \% \text{ of DL} &= (20 * 0.125 / 4.482) * 100 = 56 \% \text{ OK}
 \end{aligned}$$

STEP 6 Post-Tensioning

- ❖ Calculation of **balanced loads**
 - Lateral force from **terminated tendons**

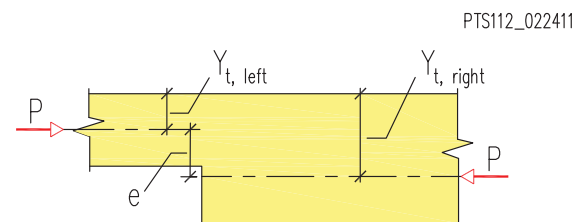


$$\begin{aligned}
 L &= 32.75 \text{ ft} ; a = 3.75 \text{ in} ; P = 26.77 \text{ k/tendon} \\
 c &= 0.20 * 32.75 = 6.55 \text{ ft} \\
 w_b &= (2 * 3 * 26.77 * 3.75 / 12) / 6.55^2 = 1.17 \text{ k/ft} \downarrow \\
 \text{Concentrated force at dead end} &= \\
 &= 2 * 3 * 26.77 * 3.75 / 12 * 6.55 = 7.66 \text{ k} \uparrow
 \end{aligned}$$

STEP 6 Post-Tensioning

- ❖ Calculation of **balanced loads up to here**
 - Lateral forced from continuous tendons
 - Lateral force from terminated tendons
 - Moments from **change in centroid** of member

- ❑ Example of force from **shift in member centroidal axis**



Moment at face of drop = M

$$M = P * \text{shift in centroid} = P * (Y_{t-Left} - Y_{t-Right})$$

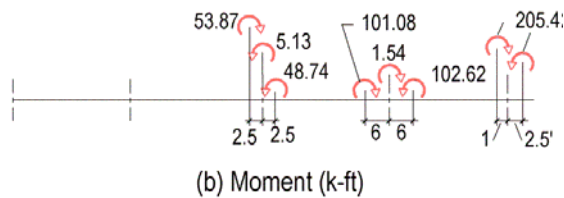
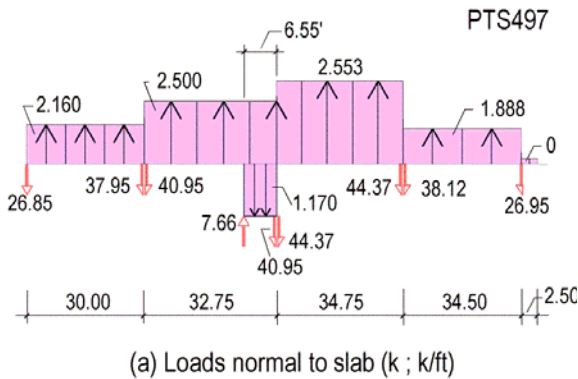
$$P = 23 * 26.77 = 615.71 \text{ k};$$

$$Y_{t-Left} = 4.75'' ; Y_{t-Right} = 5.80''$$

$$M = 615.71(4.75'' - 5.80'') / 12 = -53.87 \text{ k-ft}$$

STEP 6 Post-Tensioning

- ❖ Check balanced loads for **static equilibrium**
 - Sum of **forces** must **add up to zero**
 - Sum of **moments** must **add up to zero**
 - **Correct errors**, if equilibrium is not satisfied



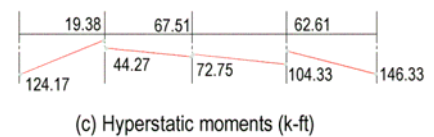
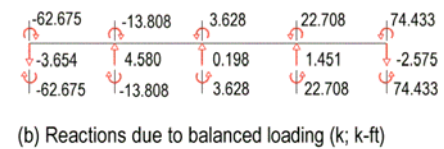
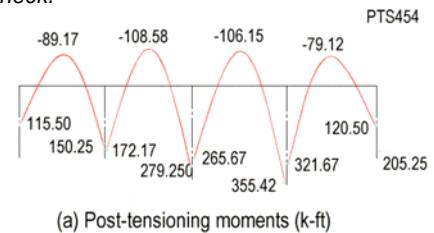
Balanced Loading

STEP 6 Post-Tensioning

- ❖ Calculate **actions** from **balanced loads**
 - Obtain moments at face-of-supports and mid-spans
 - Note the **reactions**. The reactions are **hyperstatic** forces..

Comments:

Moments will be used for **serviceability** check. Reactions will be used for **Strength** check.



Post-Tensioning Actions on Design Strip

STEP 7 Code Check for Serviceability

- ❖ ACI 318-14 requirements for serviceability
 - Stress check
 - Minimum reinforcement
 - Deflection check.
- ❖ **Load combinations**
 - **Total** (quasi permanent)
1.00DL + 1.00LL + 1.00PT
 - **Sustained** (frequent)
1.00DL + 0.30LL + 1.00PT

Stress check

Using **engineering judgment**, select the locations that are likely to be critical. Typically, these are at the **face of support** and for hand calculation at **mid-span**

At each section selected for check, use the design actions applicable to the **entire design section** and apply them to the **entire cross-section** to arrive at **code-intended hypothetical** stresses used in code check.

$$\sigma = (M_D + M_L + M_{PT})/S + P/A$$

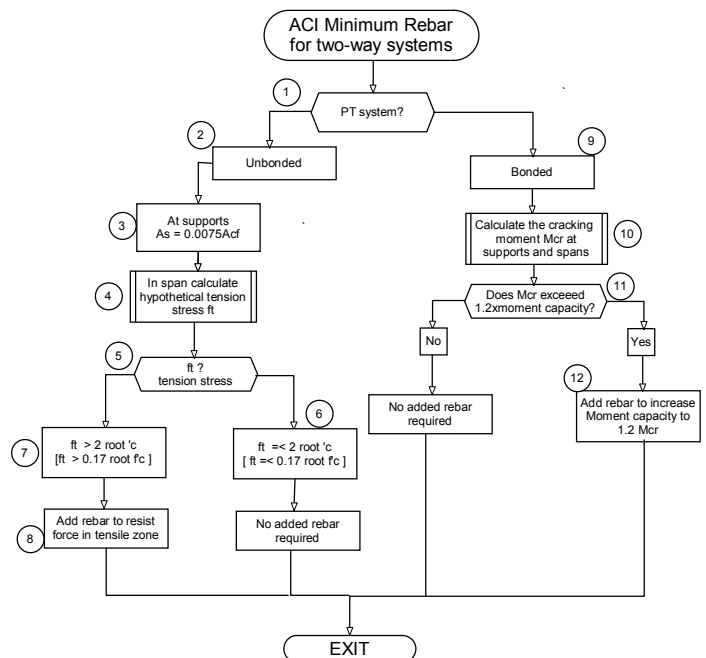
$S = I/Y_c$; I = second moment of area of ;

Y_c = distance to farthest tension fiber

STEP 6 Post-Tensioning

❖ ACI 318-14 Minimum Reinforcement

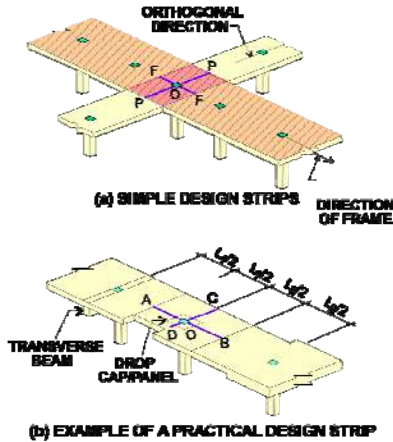
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STEP 6 Post-Tensioning

❖ ACI 318-14 Minimum Reinforcement

- Rebar **over support is function of geometry** of the design strip and the strip in the orthogonal direction
- Rebar in **span is a function** of the magnitude of the hypothetical **tensile stress**



$$A_s = 0.00075 \cdot A_{cf}$$

A_s = Area of steel required

A_{cf} = **Larger of cross-sectional area** of the strip in direction of analysis and orthogonal to it.

STEP 6 Post-Tensioning

❖ ACI 318-14 Minimum Reinforcement

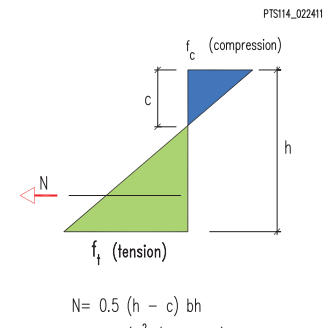
- Rebar **over support is function of geometry** of the design strip and the strip in the orthogonal direction
- Rebar in **span** is a function of the magnitude of the **hypothetical tensile stress**

In span, provide rebar if the hypothetical tensile stress **exceeds** $2\sqrt{f'_c}$

The amount of reinforcement A_s is given by:

$$A_s = N / (0.5f_y)$$

where N is the tensile force in tension zone



h = member thickness; b = design section width

STEP 7 Deflection Check

- ❖ Read deflections from the frame analysis of the design strip for dead, live and PT; (Δ_{DL} , Δ_{LL} , and Δ_{PT}).
- ❖ Make the following load combinations and check against the allowable values for each case

➤ Total Deflection

$(1 + 2)(\Delta_{DL} + \Delta_{PT} + 0.3 \Delta_{LL}) + 0.7 \Delta_{LL} < \text{span}/240$
This is on the premise of **sustained load being 0.3** time the design live load. It is for visual effects; Provide **camber** to reduce value, where needed and practical

➤ Immediate deflection from live load

$\Delta_{\text{immediate}} = 1.00 \Delta_L < \text{span}/480$
This check is applicable, where non-structural members are likely to be damaged. Otherwise, span/240 applies

➤ Presence of **members likely to be damaged** from sustained deflection

$$(1 + 2)(0.3 \Delta_{LL}) + 0.7 \Delta_{LL} < \text{span}/360$$

The above can be **exceeded**, if larger values are **acceptable** for the **specific application**

STEP 8 Strength Check

❖ Steps in strength check

- **Load combinations**
- Determination of **hyperstatic** actions
- Calculation of **design moments** (M_u)
- Calculate **capacity/rebar** for design moment M_u
- Check for **punching shear**
- Check/detail for **unbalanced moment** at support

❖ Load combinations

$$U1 = 1.2DL + 1.6LL + 1.0HYP$$

$$U2 = 1.4DL + 1.0HYP$$

where, HYP is moment due to hyperstatic actions from prestressing

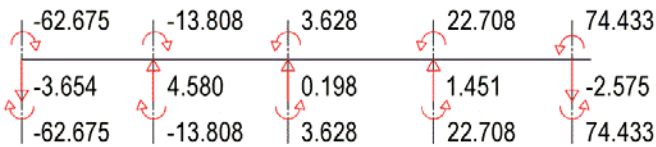
❖ Determination of **Hyperstatic** actions

- **Direct Method** – based on reactions from balanced loads
- **Indirect Method** – Using primary and post-tensioning moments

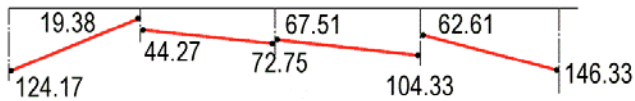
STEP 8 Strength Check

- ❖ Determination of Hyperstatic actions
 - **Direct Method** – based on **reactions** from balanced loads

PTS458



(a) Reaction due to balanced loading (k; k-ft)

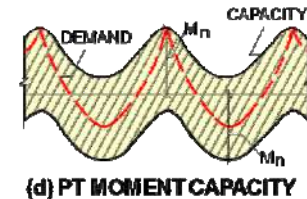
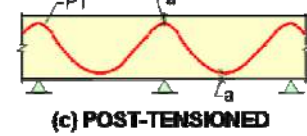
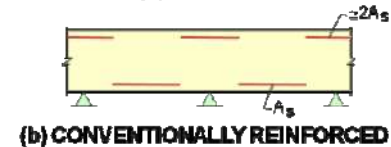
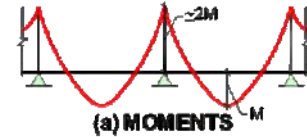


(b) Hyperstatic moments (k-ft)

Post-Tensioning Actions on Design Strip

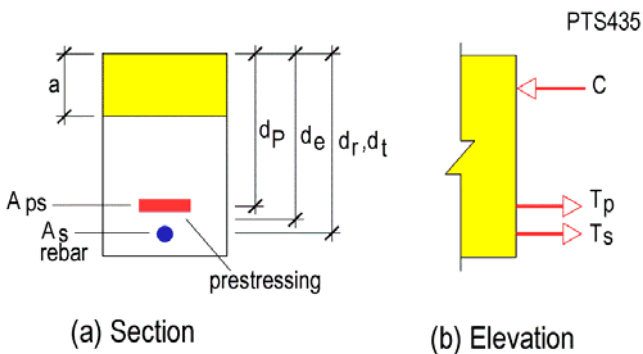
STEP 8 Strength Check

- ❖ A comment on **capacity** versus **demand**
 - Post-tensioned elements possess both **positive** and **negative moment capacity** along the **entire** element's length
 - Add rebar, where capacity falls short of demand
 - Find capacity and compare it with demand



STEP 8 Strength Check

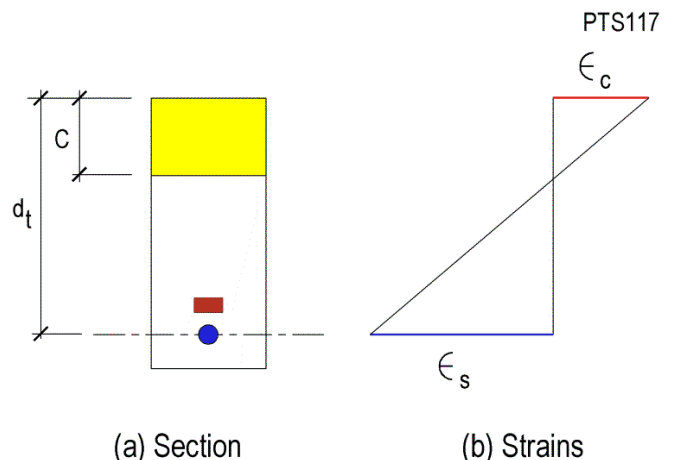
- For **capacity** calculation use the **simplified** relationship applicable to common building structures reinforced with unbonded tendons



- $f'_c \geq 4000$ psi ; $P/A \leq 250$ psi
- $c/d_t \leq 0.375$; d_t is distance from compression fiber to farthest tension rebar
- Tendon Length $\leq 125'$ for single end stressing; if length $\leq 250'$ double end stressing
- f_{ps} is conservatively 215 ksi if span is less than 35 ft
- f_{ps} is conservatively 195 ksi if span is greater than 35 ft

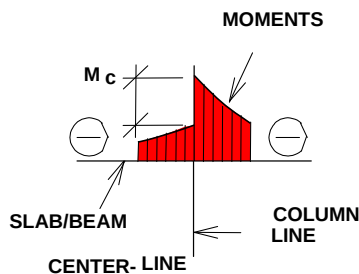
STEP 8 Strength Check

- ❖ Check for **ductility**
 - Ductility is adequate, if $c/d_t \leq 0.345$ dt. This condition guarantees that steel will yield, before concrete in compression crushes.
 - Add compression rebar if $c/d_t > 0.345$ dt



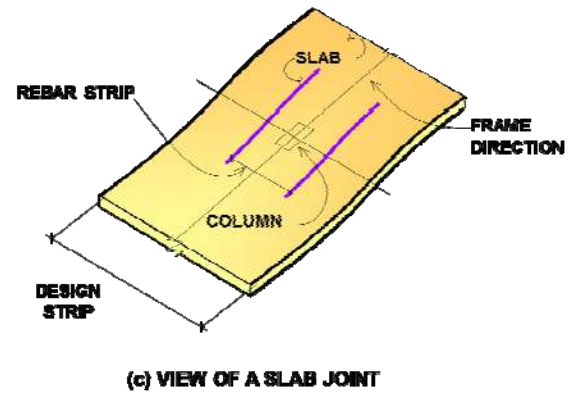
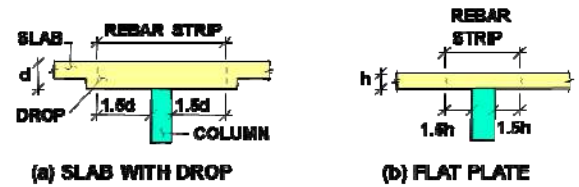
STEP 8 Strength Check

- ❖ Verify adequacy (detail) of the design for **transfer of unbalanced moment** at supports
- Unbalanced moment (M_c) is defined as the **difference** between the design moments on the **opposite sides** of a column support. This is the moment that is resisted by the support.
- The **reinforcement** associated with the transfer of unbalanced moment must be placed over a **narrow band** at the support (next slide)
- In most cases, this **provision leads to a "detailing"** requirement, as opposed to added rebar, since the reinforcement for **slab design is in excess** of that needed for transfer of unbalanced moment.



STEP 8 Strength Check

- ❖ Transfer of unbalanced moment



Place the **reinforcement** within the narrow **band** identified as rebar strip

Punching Shear Design

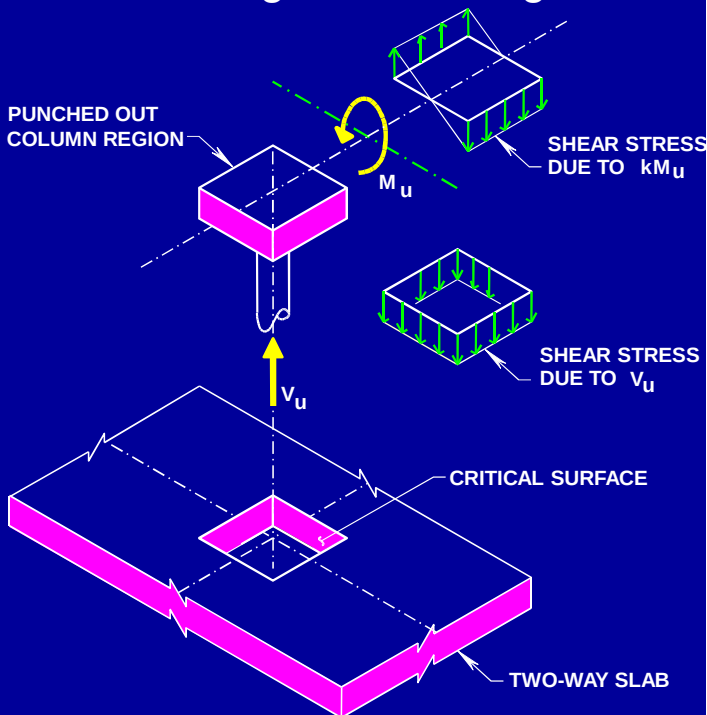


ILLUSTRATION OF CRITICAL SURFACE
FOR THE EVALUATION OF PUNCHING SHEAR STRESSES

Definition based on ACI 318

STEP 9 Check for Transfer of Prestressing

At stressing:

- ☐ Tendon has its **maximum force**;
- ☐ Concrete is at its **weakest strength**; and
- ☐ **Live load** to counteract prestressing is **absent**



STEP 9 Check for Transfer of Prestressing

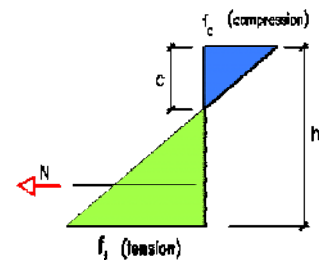
- Add **rebar** where “representative” (hypothetical) **tension stresses exceed** the allowable threshold
- Do not exceed “representative” hypothetical **compressive stresses**. **Wait** until concrete gains **adequate strength**



STEP 9 Check for Transfer of Prestressing

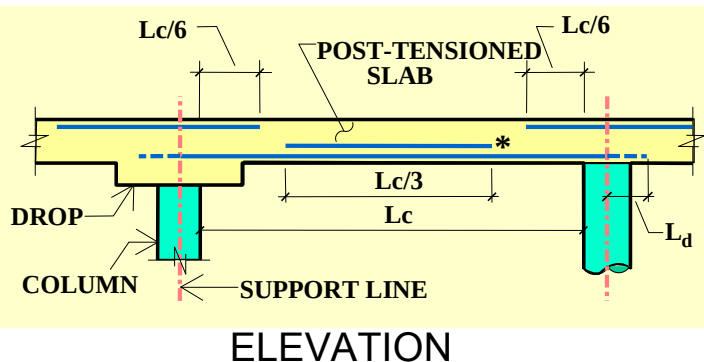
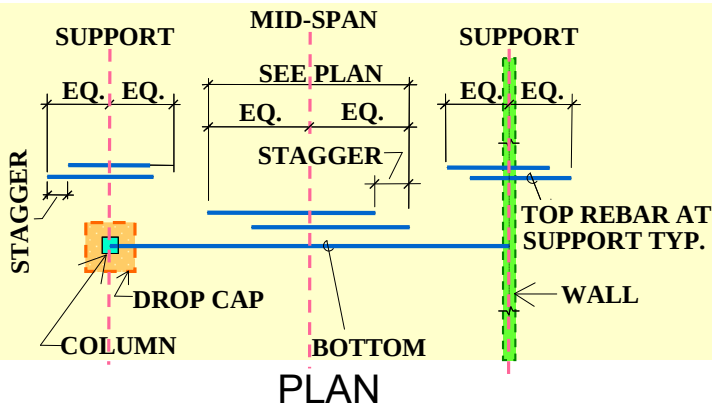
- ❖ Load combination
 - $U = 1.00 \cdot \text{Selfweight} + 1.15 \cdot \text{PT}$
- ❖ Allowable stresses
 - Tension $3\sqrt{f'_{ci}}$;
 - Compression $0.60f'_{ci}$
 - If **tension exceeds**, add **rebar** in tensile zone to resist N

$$A_s = N / (0.5 f_y)$$
- ✓ If **compression exceeds**, **wait** until concrete gains adequate strength



STEP 10 Detailing

❖ Position of rebar



Thank you for listening

QUESTIONS ?

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